

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12414285
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Jang; Semyeong et al.

Semiconductor structure and manufacturing method thereof

Abstract

The present disclosure provides a semiconductor structure and a manufacturing method thereof. The semiconductor structure includes: a base including bit lines arranged at intervals and extending along a first direction, and a semiconductor channel located on partial top surfaces of the bit lines, where along a direction from the bit line to the semiconductor channel, the semiconductor channel includes a first region, a second region, and a third region that are arranged sequentially; a dielectric layer located between adjacent two of the bit lines and on a sidewall of the semiconductor channel; a gate structure at least surrounding the dielectric layer in the second region and extending along a second direction, where the first direction is different from the second direction; an electrical connection layer covering a top surface of the third region and extending to a partial sidewall of the semiconductor channel.

Inventors: Jang; Semyeong (Hefei, CN), Moon; Joonsuk (Hefei, CN), Xiao; Deyuan (Hefei, CN), Chin; Jo-Lan (Hefei, CN), Hong; Minki (Hefei, CN)

Applicant: CHANGXIN MEMORY TECHNOLOGIES, INC. (Hefei, CN)

Family ID: 88415267

Assignee: CHANGXIN MEMORY TECHNOLOGIES, INC. (Hefei, CN)

Appl. No.: 17/932418

Filed: September 15, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230345698 A1	Oct. 26, 2023

Foreign Application Priority Data

CN	202210451692.6	Apr. 26, 2022
----	----------------	---------------

Related U.S. Application Data

continuation parent-doc WO PCT/CN2022/111403 20220810 PENDING child-doc US 17932418

Publication Classification

Int. Cl.: H10B12/00 (20230101)

U.S. Cl.:

CPC H10B12/05 (20230201); H10B12/315 (20230201);

Field of Classification Search

CPC: H10B (12/05); H10B (12/315); H10B (12/31)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7478294	12/2008	Rong et al.	N/A	N/A
8502186	12/2012	Chung	N/A	N/A
9147595	12/2014	Lee et al.	N/A	N/A
10181472	12/2018	Huang	N/A	N/A
10388659	12/2018	Zhang	N/A	N/A
10546862	12/2019	Yang et al.	N/A	N/A
11069679	12/2020	Wu et al.	N/A	N/A
2008/0251825	12/2007	Lee	N/A	N/A
2011/0012085	12/2010	Deligianni	257/E21.409	H10D 30/6735
2011/0121396	12/2010	Lee	257/E29.264	G11C 11/403
2013/0234240	12/2012	Moon et al.	N/A	N/A
2013/0323920	12/2012	Chang	438/586	H10B 12/01
2016/0247892	12/2015	Masuoka et al.	N/A	N/A
2018/0248018	12/2017	Park et al.	N/A	N/A
2019/0035795	12/2018	Boemmels	N/A	N/A
2019/0067459	12/2018	Balakrishnan et al.	N/A	N/A
2022/0037483	12/2021	Lowe	N/A	G11C 11/4023
2022/0068922	12/2021	Chen et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
100444287	12/2007	CN	N/A
101897008	12/2009	CN	N/A
102779828	12/2011	CN	N/A

103904115	12/2013	CN	N/A
109216278	12/2018	CN	N/A
107039084	12/2019	CN	N/A
109712976	12/2020	CN	N/A
113838802	12/2020	CN	N/A
114141712	12/2021	CN	N/A
114203806	12/2021	CN	N/A
114784006	12/2021	CN	N/A

OTHER PUBLICATIONS

International Search Report cited in PCT/CN2022/109754 mailed Jan. 17, 2023, 9 pages. cited by applicant

International Search Report cited in PCT/CN2022/111403 mailed Jan. 16, 2023, 10 pages. cited by applicant

International Search Report cited in PCT/CN2022/098253 mailed Nov. 30, 2022, 9 pages. cited by applicant

Non-final office action issued in U.S. Appl. No. 17/934,655 mailed on Mar. 28, 2025, 9 pages. cited by applicant

Non-final office action issued in U.S. Appl. No. 17/893,058 mailed on Mar. 14, 2025, 10 pages. cited by applicant

Primary Examiner: Fletcher; Marlon T

Assistant Examiner: Smith; Samuel Jonathan

Attorney, Agent or Firm: Kilpatrick Townsend & Stockton LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This is a continuation of International Application No. PCT/CN2022/111403, filed on Aug. 10, 2022, which claims the priority to Chinese Patent Application No. 202210451692.6, titled “SEMICONDUCTOR STRUCTURE AND MANUFACTURING METHOD THEREOF” and filed with the China National Intellectual Property Administration (CNIPA) on Apr. 26, 2022. The entire contents of International Application No. PCT/CN2022/111403 and Chinese Patent Application No. 202210451692.6 are incorporated herein by reference.

TECHNICAL FIELD

(1) The present disclosure relates, but is not limited, to a semiconductor structure and a manufacturing method thereof.

BACKGROUND

(2) With the increasingly high integration density of the dynamic memory, while the arrangement manners of transistors in a dynamic memory array structure and the way to reduce the size of individual functional devices in the dynamic memory array structure are studied, it is necessary to consider the effect of small-scale functional devices on the overall electrical performance of the semiconductor structure.

(3) When vertical gate-all-around (GAA) transistors are used as access transistors of the dynamic memory, the area occupied by the GAA transistors can reach $4F_{\text{sup.2}}$ (F : the smallest pattern size available under given process conditions). In principle, this arrangement can improve the density

and efficiency, but higher demands are placed on the electrical connection accuracy and conductive performance between small-sized functional devices, and higher demands are also placed on the overall electrical performance of the semiconductor structure.

SUMMARY

(4) An overview of the subject described in detail in the present disclosure is provided below, which is not intended to limit the protection scope of the claims.

(5) The present disclosure provides a semiconductor structure and a manufacturing method thereof.

(6) A first aspect of the present disclosure provides a semiconductor structure, including: a base including bit lines arranged at intervals and extending along a first direction, and a semiconductor channel located on partial top surfaces of the bit lines, where along a direction from the bit line to the semiconductor channel, the semiconductor channel includes a first region, a second region, and a third region that are arranged sequentially; a dielectric layer located between adjacent two of the bit lines and on a sidewall of the semiconductor channel; a gate structure at least surrounding the dielectric layer in the second region and extending along a second direction, where the first direction is different from the second direction; an electrical connection layer covering a top surface of the third region and extending to a partial sidewall of the semiconductor channel, where the electrical connection layer and the dielectric layer cover a surface of the semiconductor channel; and an insulating layer located between adjacent two of the semiconductor channels on a same bit line and isolating the gate structure as well as the dielectric layer on adjacent two of the dielectric layers.

(7) A second aspect of the present disclosure provides a method of manufacturing a semiconductor structure, including: providing a base, where the base includes bit lines arranged at intervals and extending along a first direction, and a semiconductor channel located on partial top surfaces of the bit lines, and along a direction perpendicular to the top surface of the bit line, the semiconductor channel includes a first region, a second region, and a third region that are arranged sequentially; forming a dielectric layer, where the dielectric layer is located between adjacent two of the bit lines and on a sidewall of the semiconductor channel; forming a gate structure, where the gate structure at least surrounds the dielectric layer in the second region and extends along a second direction, and the first direction is different from the second direction; forming an electrical connection layer, where the electrical connection layer covers a top surface of the third region and extends to a partial sidewall of the semiconductor channel, and the electrical connection layer and the dielectric layer cover a surface of the semiconductor channel; and forming an insulating layer, where the insulating layer is located between adjacent two of the semiconductor channels on a same bit line.

(8) Other aspects of the present disclosure are understandable upon reading and understanding of the accompanying drawings and detailed description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings incorporated into the specification and constituting a part of the specification illustrate the embodiments of the present disclosure, and are used together with the description to explain the principles of the embodiments of the present disclosure. In these accompanying drawings, similar reference numerals are used to represent similar elements. The accompanying drawings in the following description are some rather than all of the embodiments of the present disclosure. Those skilled in the art may obtain other accompanying drawings based on these accompanying drawings without creative efforts.

(2) FIG. 1 is a schematic top view of a semiconductor structure according to an embodiment of the present disclosure;

(3) FIG. 2 is a schematic cross-sectional view of the semiconductor structure in FIG. 1 along a first

cross-sectional direction AA1;

(4) FIG. 3 is another schematic cross-sectional view of the semiconductor structure in FIG. 1 along the first cross-sectional direction AA1;

(5) FIG. 4 is a schematic cross-sectional view of the semiconductor structure in FIG. 1 along a second cross-sectional direction BB1;

(6) FIG. 5 is a schematic cross-sectional view in which in the semiconductor structure in FIG. 1, a first gate layer surrounds a semiconductor channel, and a second gate layer surrounds the semiconductor channel; and

(7) FIG. 6 to FIG. 18 are schematic structural cross-sectional views corresponding to various steps of a method of manufacturing a semiconductor structure according to another embodiment of the present disclosure.

DETAILED DESCRIPTION

(8) The technical solutions in the embodiments of the present disclosure are described below clearly and completely with reference to the accompanying drawings in the embodiments of the present disclosure. Apparently, the described embodiments are merely some rather than all of the embodiments of the present disclosure. All other embodiments obtained by those skilled in the art based on the embodiments of the present disclosure without creative efforts should fall within the protection scope of the present disclosure. It should be noted that the embodiments in the present disclosure and features in the embodiments may be combined with each other in a non-conflicting manner.

(9) As mentioned in the background, the electrical performance of the semiconductor structure in the prior art needs to be improved.

(10) After analysis, it is found that in the current GAA transistor structure, the conduction or turn-off of the semiconductor channel is controlled by applying a voltage to the gate structure and the third region of the semiconductor channel. The third region of the semiconductor channel may receive an externally supplied voltage through the electrical connection layer. However, to achieve higher integration density, the sizes of both the gate structure and the semiconductor channel are further reduced, and the contact area between the electrical connection layer and the third region of the semiconductor channel is further reduced, which increase the difficulty of voltage control of semiconductor channels, thus reducing the electrical property of the semiconductor structure.

(11) Embodiments of the present disclosure provide a semiconductor structure and a manufacturing method thereof. In the semiconductor structure, vertical GAA transistors are formed in a base, and a 3D stacked semiconductor structure can be formed, which are beneficial to improving the integration density of the semiconductor structure. In addition, the electrical connection layer covers the top surface of the third region of the semiconductor channel and extends to a partial sidewall of the semiconductor channel, which therefore is beneficial to increasing the contact area between the electrical connection layer and the semiconductor channel, reducing the contact resistance between the electrical connection layer and the third region of the semiconductor channel, and improving the transmission performance of electrical signals between the electrical connection layer and the third region of the semiconductor channel, thereby improving electrical properties of the semiconductor structure. Further, this is beneficial to improving the transmission performance of electrical signals between the other conductive structures and the third region of the semiconductor channel, when the conductive structures of other functional devices are subsequently formed on the electrical connection layer. Therefore, the embodiments of the present disclosure are beneficial to improving the ability of the electrical signal that the electrical connection layer transmits to the semiconductor channel to control the semiconductor channel, thereby improving the on/off sensitivity of the GAA transistor.

(12) A first embodiment of the present disclosure provides a semiconductor structure. The semiconductor structure provided by the first embodiment of the present disclosure is described in detail below with reference to the drawings. FIGS. 1 to 5 are schematic structural views of the

semiconductor structure according to an embodiment of the present disclosure. FIG. 1 is a schematic top view of a semiconductor structure according to an embodiment of the present disclosure; FIG. 2 is a schematic cross-sectional view of the semiconductor structure in FIG. 1 along a first cross-sectional direction AA1; FIG. 3 is another schematic cross-sectional view of the semiconductor structure in FIG. 1 along the first cross-sectional direction AA1; FIG. 4 is a schematic cross-sectional view of the semiconductor structure in FIG. 1 along a second cross-sectional direction BB1; and FIG. 5 is a schematic cross-sectional view in which in the semiconductor structure in FIG. 1, a first gate layer surrounds a semiconductor channel, and a second gate layer surrounds the semiconductor channel.

(13) With reference to FIGS. 1 to 5, the semiconductor structure includes: a base **100** including bit lines **101** arranged at intervals and extending along a first direction X, and a semiconductor channel **102** located on partial top surfaces of the bit lines **101**, where along a direction Z from the bit lines **101** to the semiconductor channels **102**, the semiconductor channel **102** includes a first region I, a second region II, and a third region III that are arranged sequentially; a dielectric layer **103** located between adjacent two of the bit lines **101** and on a sidewall of the semiconductor channel **102**; a gate structure **104** at least surrounding the dielectric layer **103** in the second region II and extending along a second direction Y, where the first direction X is different from the second direction Y; an electrical connection layer **105** covering a top surface of the third region III and extending to a partial sidewall of the semiconductor channel **102**, where the electrical connection layer **105** and the dielectric layer **103** cover a surface of the semiconductor channel **102**; and an insulating layer **106** located between adjacent two of the semiconductor channels **102** on a same bit line **101** and isolating the gate structure **104** as well as the dielectric layer **103** on adjacent two of the dielectric layers **103**.

(14) With reference to FIG. 2, the semiconductor channel **102**, the dielectric layer **103** surrounding the sidewall of the semiconductor channel **102**, and the gate structure **104** form a vertical GAA transistor. The base **100** includes a substrate **110**, and the bit line **101** is located between the substrate **110** and the GAA transistor, such that a 3D stacked semiconductor structure can be formed, which is beneficial to improving the integration density of the semiconductor structure.

(15) It should be noted that, with reference to FIG. 2, both the first region I and the third region III can be used as the source or drain of the GAA transistor, and the gate structure **104** is configured to control the turn-on/off of the GAA transistor.

(16) In some embodiments, with reference to FIG. 1, the first direction X is perpendicular to the second direction Y, such that the semiconductor channels **102** are arranged in a 4F_{sup.2} manner (F: the smallest pattern size available under given process conditions), which is beneficial to improving the integration density of the semiconductor structure. In practical applications, the first direction X only needs to intersect the second direction Y, and the angle between the two may not be 90°.

(17) With reference to FIG. 2, the electrical connection layer **105** not only covers the top surface of the third region III of the semiconductor channel **102**, but also extends to a partial sidewall of the semiconductor channel **102**, which therefore is beneficial to increasing the contact area between the electrical connection layer **105** and the third region III, reducing the contact resistance between the electrical connection layer **105** and the third region III, and improving the transmission performance of electrical signals between the electrical connection layer **105** and the third region III, thereby improving electrical properties of the semiconductor structure.

(18) It should be noted that, with reference to FIG. 2, a plurality of spaced-apart bit lines **101** may be formed in the base **100**, and the bit lines **101** each may be in contact with at least one first region I. FIGS. 1 to 4 show four spaced-apart bit lines **101**, where the bit lines **101** each are in contact with four first regions I. In the practical application, the quantity of the bit lines **101** and the quantity of the first regions I in contact with each of the bit lines **101** may be set properly according to actual electrical requirements.

(19) The embodiments of the present disclosure are further described in detail below with reference

to FIGS. 1 to 5.

(20) In some embodiments, with reference to FIG. 2, the base **100** may be made of an elemental semiconductor material or a crystalline inorganic compound semiconductor material. The elemental semiconductor material may include silicon or germanium, and the crystalline inorganic compound semiconductor material may include silicon carbide, silicon germanium, gallium arsenide or indium gallium.

(21) In some embodiments, with reference to FIG. 2, the base **100** includes a bit line **101** and a semiconductor channel **102**. If the base **100**, the bit lines **101** and the semiconductor channels **102** have the same semiconductor element, the semiconductor channel **102** and the bit line **101** may be formed by using the same film structure. The film structure is composed of the semiconductor element, such that the semiconductor channels **102** and the bit lines **101** are integrated, thereby avoiding an interface state defect between the semiconductor channels **102** and the bit lines **101** and improving the performance of the semiconductor structure.

(22) The semiconductor element may include at least one selected from the group consisting of silicon, carbon, germanium, arsenic, gallium and indium. Subsequent description is made by using an example in which both the bit line **101** and the semiconductor channel **102** both include the silicon element.

(23) In some embodiments, with reference to FIG. 2, the semiconductor channel **102** may include doping elements, which is beneficial to improving the conductivity of the semiconductor channel **102**. Therefore, this is beneficial to reducing the turn-on voltage between the first region I and the third region III, that is, the turn-on voltage between the source and the drain of the GAA transistor. The doping element is a P-type doping element or an N-type doping element. For example, the N-type doping element may be at least one of arsenic, phosphorus, or antimony; and the P-type doping element may be at least one of boron, indium, or gallium.

(24) In some embodiments, the GAA transistor may be a junctionless transistor, that is, the doping elements in the first region I, the second region II, and the third region III are of the same type. The “junctionless” here refers to no PN junction, that is, the first region I, the second region II, and the third region III have the same concentration. In this way, there is no need to perform additional doping in the first region I and the third region III, thereby avoiding the problem that the doping process in the first region I and the third region III is difficult to control. Especially as the size of the transistor is further reduced, if the first region I and the third region III are additionally doped, the doping concentration will become more difficult to control. In addition, because the device is a junctionless transistor, it avoids the use of an ultra-steep source/drain doping process to make an ultra-steep PN junction in a nanoscale range. Therefore, problems such as threshold voltage drift and leakage current increase caused by abrupt changes in the doping concentration can be avoided, and the short channel effect can be suppressed. Such a design further improves the integration density and electrical performance of the semiconductor structure. Understandably, the additional doping herein refers to doping to make the types of the doping elements in the first region I and the third region III different from that of the doping element in the second region II.

(25) In some embodiments, with reference to FIG. 3 and FIG. 4, the dielectric layer **103** may include: a first dielectric layer **113** located between adjacent bit lines **101** and between the semiconductor channels **102** of the first regions I on the adjacent bit lines **101**; the second dielectric layer **123** located on the sidewall of the semiconductor channel **102** in the first region and the sidewall of the first dielectric layer **113** in the first region I; the third dielectric layer **133** surrounding the sidewall of the semiconductor channel **102** in the second region II; a fourth dielectric layer **143** located between the third dielectric layer **133** and the insulating layer **106** and spaced between the first gate layer **114** and the second gate layer **124**; and the fifth dielectric layer **153**, where the fifth dielectric layer and the electrical connection layer **105** cover the surface of the semiconductor channel **102** in the third region III.

(26) With reference to FIGS. 3 and 4, the first dielectric layer **113** between adjacent bit lines **101** is

configured to electrically insulate adjacent bit lines **101**. The first dielectric layer **113**, the second dielectric layer **123** and the insulating layer **106** between the semiconductor channels **102** of the first regions I on adjacent bit lines **101** work together, to electrically insulate the semiconductor channels **102** in the first region I spaced along the first direction X and/or the second direction Y. The third dielectric layer **133** surrounding the sidewall of the semiconductor channel **102** in the second region II may be configured to isolate the first gate layer **114** from the semiconductor channel **102** in the second region II. The third dielectric layer **133** and the fourth dielectric layer **143** may be configured to jointly isolate the first gate layer **114** and the second gate layer **124**. The fifth dielectric layer **153** surrounding the surface of the semiconductor channel **102** in the third region III and the insulating layer **106** work together, to electrically insulate the semiconductor channels **102** in the third region III spaced along the first direction X and/or the second direction Y. (27) In some embodiments, with reference to FIG. 3, the third dielectric layer **133** may further be located on a partial sidewall of the second dielectric layer **123**, which is beneficial to further ensuring the insulation between the first gate layer **114** and the semiconductor channel **102**. The fifth dielectric layer **153** may further be located on a partial sidewall of the third dielectric layer **133**, which is beneficial to further ensuring the insulation between the second gate layer **124** and the semiconductor channel **102**.

(28) In some embodiments, with reference to FIGS. 3 and 4, the first dielectric layer **113**, the second dielectric layer **123**, the third dielectric layer **133**, the fourth dielectric layer **143**, and the fifth dielectric layer **153** may be made of a same material, such as, silicon oxide. In another embodiment, the five dielectric layers may be made of different materials, provided that these materials have good insulating effects.

(29) The specific structure of the gate structure **104** is described in detail below through two embodiments.

(30) In some embodiments, with reference to FIG. 2, the gate structure **104** surrounds the dielectric layer **103** of the second region II and extends along the second direction Y (with reference to FIG. 1).

(31) In other embodiments, with reference to FIGS. 3 and 4, the gate structure **104** includes: a first gate layer **114** surrounding the dielectric layer **103** in the second region II and extending along the second direction Y (with reference to FIG. 1); and the second gate layer **124** surrounding the dielectric layer **103** of the third region III, and in the direction Z perpendicular to the top surface of the bit line **101**, where the second gate layer **124** is spaced from the first gate layer **114**.

(32) In this way, the first gate layer **114** and the second gate layer **124** are designed to control the same semiconductor channel **102**, such that the capabilities of the first gate layer **114** and the second gate layer **124** to control the semiconductor channel **102** complement each other. When the lack of turn-off capability of the semiconductor channel **102** in one of the gate layers results in a large leakage current, such as GIDL (gate-induced drain leakage), the semiconductor channel **102** is turned off by using another gate layer, thereby ensuring good capabilities of the first gate layer **114** and the second gate layer **124** to control the semiconductor channel **102** on the whole. This is beneficial to reducing the leakage current in the semiconductor structure, to improve the overall electrical performance of the semiconductor structure.

(33) The specific situation that the gate structure **104** includes the first gate layer **114** and the second gate layer **124** is described in detail below with reference to FIGS. 3 to 5.

(34) In some embodiments, with reference to FIG. 3, in a plane perpendicular to the sidewall of the semiconductor channel **102**, the first gate layer **114** surrounds the semiconductor channel **102**, that is, the second region II has a first cross section. The second gate layer **124** surrounds the semiconductor channel **102**, that is, the third region III has a second cross section. The area of the first cross section is greater than that of the second cross section. In this way, the cross-sectional area of the third region III is smaller, which is beneficial to improving the ability of the second gate layer **124** to control the semiconductor channel **102**. In other words, the smaller the threshold

voltage required, the easier it is to turn on or off the GAA transistor. It is further beneficial to adjusting the ability of the second gate layer **124** to control the semiconductor channel **102** to compensate for the instability of the ability of the first gate layer **114** to control the semiconductor channel **102**, thereby ensuring the overall controllability of the semiconductor channel **102**, to improve the overall electrical performance of the semiconductor structure.

(35) The ratio of the area of the first cross section to that of the second region may be 1.5 to 2.5.

(36) In some embodiments, with reference to FIG. **3**, in a plane perpendicular to the sidewall of the semiconductor channel **102**, a cross-sectional area of the semiconductor channel **102** in the first region I, a cross-sectional area of the semiconductor channel **102** in the second region II, and the cross-sectional area of the semiconductor channel **102** in the third region III are decreased sequentially.

(37) With reference to FIG. **3**, along the first direction X, the side surface of the second gate layer **124** away from the sidewall of the semiconductor channel **102** is flush with that of the first gate layer **114** away from the sidewall of the semiconductor channel **102**. In this way, it is beneficial to further increasing the overall volume of the second gate layer **124**, and improving the conductivity of the second gate layer **124**, thereby improve the ability of the second gate layer **124** to receive electrical signals, such that on/off of the semiconductor channel **102** can be controlled by applying a smaller voltage to the second gate layer **124**, to improve the on/off sensitivity of GAA transistors.

(38) In some embodiments, with reference to FIGS. **1** to **4**, in directions perpendicular to the sidewalls of the semiconductor channel **102**, that is, along the first direction X and the second direction Y, the first gate layer **114** is thinner than the second gate layer **124**. Along the direction Z from the bit line **101** to the semiconductor channel **102**, the first gate layer **114** is higher than the second gate layer **124**. In this way, it is beneficial to properly using the gap between adjacent semiconductor channels **102** to design the dimensions of the first gate layer **114** and the second gate layer **124**, to reduce the difference between the volume of the first gate layer **114** and the volume of the second gate layer **124**, such that both the first gate layer **114** and the second gate layer **124** each have a suitable volume. This is beneficial to reducing the difference between the conductivity of the first gate layer **114** and the conductivity of the second gate layer **124**, to reduce the difference in the abilities of the first gate layer **114** and the second gate layer **124** to control the semiconductor channel **102**, thereby ensuring good ability of the first gate layer **114** and the second gate layer **124** to control the semiconductor channel **102** on the whole, reducing the leakage current in semiconductor structures, and improving the overall electrical performance of the semiconductor structure.

(39) In addition, with reference to FIG. **3**, along the direction Z, the first gate layer **114** may be higher than the second gate layer **124**, which is beneficial to ensuring a larger surrounding area of the semiconductor channel **102** by the first gate layer **114**, and improving the ability of the first gate layer **114** to control the semiconductor channel **102**. The ratio of the length of the first gate layer **114** to that of the second gate layer **124** may be 1.5 to 4.

(40) In some embodiments, with reference to FIG. **4**, along the direction Z from the bit line **101** to the semiconductor channel **102**, a ratio of the height of the first gate layer **114** surrounding the sidewall of the second region II to that of the semiconductor channel **102** of the second region II is 1/2 to 9/10. In this way, a ratio of the height of the third dielectric layer **133** covering the sidewall of the semiconductor channel **102** in the second region II to that of the first gate layer **114** is 1/2 to 9/10, such that there is an appropriate distance between the first gate layer **114** and the top surface of the semiconductor channel **102** in the second region II, and along the direction Z, there is an appropriate distance between the first gate layer **114** and the second gate layer **124**, to reduce the parasitic effect between the first gate layer **114** and the second gate layer **124**, thereby improving the electrical performance of the semiconductor structure.

(41) In some embodiments, with reference to FIG. **4**, along the direction Z from the bit line **101** to the semiconductor channel **102**, a ratio of the height of the second gate layer **124** surrounding the

sidewall in the third region III to that of the semiconductor channel **102** in the third region III is $1/2$ to $9/10$. In this way, it is beneficial to having an appropriate distance between the second gate layer **124** and the top surface of the semiconductor channel **102** in the third region III, such that along the direction Z, there is an appropriate distance between the second gate layer **124** and the electrical connection layer **105**, to reduce the parasitic effect between the second gate layer **124** and the electrical connection layer **105**, thereby improving the electrical performance of the semiconductor structure.

(42) In some embodiments, with reference to FIG. 5, a single first gate layer **114** extends along the second direction Y and surrounds adjacent semiconductor channels **102** on adjacent bit lines **101**; and a single second gate layer **124** extends along the second direction Y and surrounds adjacent semiconductor channels **102** on adjacent bit lines **101**.

(43) With reference to FIG. 5, both materials of the first gate layer **114** and the material of the second gate layer **124** may include at least one selected from the group consisting of polysilicon, titanium nitride, tantalum nitride, copper or tungsten. In some embodiments, the second gate layer **124** may include at least one of lanthanides and/or zirconium. In this way, it is beneficial to improving the electrical performance of the second gate layer **124**, thereby improving the ability of the second gate layer **124** to control the semiconductor channel **102**. In addition, at least one of lanthanides and/or zirconium are/is doped into the second gate layer **124** to improve the conductivity of the second gate layer **124**. This is beneficial to further avoiding the conductivity difference between the first gate layer **114** and the second gate layer **124**, and reducing the difference in the abilities of the first gate layer **114** and the second gate layer **124** to control the semiconductor channel **102**, thereby improving the overall stability of the semiconductor structure.

(44) In some embodiments, with reference to FIG. 4, the vertical distance between the top surface of the first gate layer **114** away from the bit line **101** and the bottom surface of the second gate layer **124** close to the bit line **101** may be 20 nm to 60 nm. In this way, a large parasitic capacitance is avoided between the first gate layer **114** and the second gate layer **124**, mutual interference between the first gate layer **114** and the second gate layer **124** is reduced. This is beneficial to ensuring good abilities of the first gate layer **114** and the second gate layer **124** to control the semiconductor channel **102**.

(45) The electrical connection layer **105** in the semiconductor structure is described in detail below with reference to FIGS. 1 to 5.

(46) In some embodiments, with reference to FIG. 2, the electrical connection layer **105** include at least one of lanthanides and/or zirconium. In this way, the electrical performance of the electrical connection layer **105** is improved, thereby improving the transmission effect of electrical signals in the electrical connection layer **105**, to improve the ability of the electrical signal transmitted to the semiconductor channel **102** through the electrical connection layer **105** to control the semiconductor channel **102**. This is beneficial to improving the on/off sensitivity of the GAA transistor.

(47) In some embodiments, with reference to FIG. 2, along a direction Z from the bit line **101** to the semiconductor channel **102**, the electrical connection layer **105** on the top surface of the third region III is 10 nm to 50 nm. In this way, while having good conductivity, the electrical connection layer **105** has an appropriate size, thereby avoiding taking up too much space in the semiconductor structure.

(48) In some embodiments, with reference to FIGS. 3 and 4, along the directions perpendicular to the sidewalls of the semiconductor channel **102**, that is, in the first direction X and the second direction Y, the ratio of the width of the electrical connection layer **105** located on the sidewall of the semiconductor channel **102** to that of the semiconductor channel **102** in the third region III is $1/5$ to $3/5$. In an example, along a direction perpendicular to the sidewall of the semiconductor channel **102**, the width of the electrical connection layer **105** located on the sidewall of the semiconductor channel **102** may be 10 nm to 50 nm.

(49) In some embodiments, with reference to FIG. 3, along the direction Z from the bit line **101** to the semiconductor channel **102**, a ratio of the height of the electrical connection layer **105** on the sidewall the semiconductor channel **102** to that of the semiconductor channel **102** in the third region III is 1/5 to 3/5. In an example, along the direction Z, the height of the electrical connection layer **105** located on the sidewall of the semiconductor channel **102** may be 10 nm to 50 nm.

(50) In some embodiments, with reference to FIG. 3, orthographic projection of the electrical connection layer **105** on the bit line **101** is not overlapped or partially overlapped with that of the gate structure **104** on the bit line **101**. In an example, the orthographic projection of the electrical connection layer **105** on the bit line **101** is not overlapped with that of the gate structure **104** on the bit line **101**. This is beneficial to avoiding that the electrical connection layer **105** faces the gate structure **104** in the direction Z, to reduce the mutual interference between the electrical connection layer **105** and the gate structure **104**, such as a parasitic capacitance between the electrical connection layer **105** and the gate structure **104**, thereby improving the electrical performance of the semiconductor structure. In another example, the orthographic projection of the electrical connection layer **105** on the bit line **101** is partially overlapped with that of the gate structure **104** on the bit line **101**. This is beneficial to reducing the facing area between the electrical connection layer **105** and the gate structure **104**, to reduce the mutual interference between the electrical connection layer **105** and the gate structure **104**, thereby improving the electrical property of the semiconductor structure.

(51) In some embodiments, with reference to FIG. 2, the semiconductor structure further includes a metal-semiconductor compound structure **111** at least located in a bit line opposite to a bottom of the insulating layer **106**.

(52) With reference to FIG. 2, the metal-semiconductor compound structure **111** have a smaller resistivity than an unmetallized semiconductor material. Therefore, compared to the semiconductor channel **102**, the bit line **101** including the metal-semiconductor compound structure **111** has a smaller resistivity. This reduces the resistance of the bit lines **101** and the contact resistance between the bit lines **101** and the semiconductor channel **102** of the first region I, thereby further improving the electrical performance of the semiconductor structure. In addition, the resistivity of the bit line **101** may further be smaller than that of the substrate **110**.

(53) It should be noted that in some embodiments, with reference to FIG. 2, a region of the bit line **101** under the first doped region I may be made of a semiconductor material, and a partial region of the bit line **101** that is not covered by the first region I is made of a metal semiconductor compound. It may be understood that as the size of the device continues to shrink and the manufacturing process parameters are adjusted, a partial region of the bit line **101** under the first region I is made of a semiconductor material, and the remaining region of the bit line **101** under the first region I may also be made of a metal semiconductor compound. The “remaining region” here is located on a periphery of the “partial region”.

(54) For example, with reference to FIG. 2, the plurality of metal-semiconductor compound structures **111** in the bit line **101** communicate with each other to form a part of the bit line **101**. Moreover, the metal-semiconductor compound structure **111** may be partially located in the bit line **101** and partially located in the semiconductor channel **102** in the first region I. In other embodiments, a plurality of metal-semiconductor compound structures in the same bit line may be spaced from each other.

(55) In FIG. 2, the region of the base **100** defined by a dotted ellipse is a metal-semiconductor compound structure **111**. In the practical application, the size of the contact region between adjacent metal-semiconductor compound structures **111** is not limited. In other embodiments, the entire bit line may be used as the metal-semiconductor compound structure **111**.

(56) In some embodiments, with reference to FIG. 2, along a direction from the semiconductor channels **102** at two sides of the insulating layer **106** to the insulating layer **106**, that is, in the directions C1 and C2, the metal-semiconductor compound structure **111** becomes deeper.

(57) An example in which the semiconductor element is silicon is made below. The material of the metal-semiconductor compound structure **111** includes at least one selected from the group consisting of cobalt silicide, nickel silicide, molybdenum silicide, titanium silicide, tungsten silicide, tantalum silicide and platinum silicide.

(58) In some embodiments, with reference to FIG. 2, the insulating layer **106** includes: a first insulating layer **116** located between the dielectric layer **103** between adjacent semiconductor channels **102** and the gate structure **104** and extending along the second direction Y, where the top surface of the first insulating layer **116** is not lower than that of the semiconductor channel **102** in the third region III; and a second insulating layer **126** located on the top surface of the second gate layer **124** away from the bit line **101** and between the first insulating layer **116** and the electrical connection layer **105**.

(59) With reference to FIG. 2, the first insulating layer **116** and the second insulating layer **126** work together to electrically insulate adjacent semiconductor channels **102** along the first direction X, adjacent first gate layers **114**, and adjacent second gate layers **124**. In addition, the second insulating layer **126** on the top surface of the second gate layer **124** can electrically insulate the second gate layer **124** from another conductive structure.

(60) In some embodiments, with reference to FIG. 2, the first insulating layer **116** and the second insulating layer **126** may be made of a same material, such as silicon nitride. In another embodiment, the two insulating layers may be made of different materials, provided that these materials have good insulating effects. In the same etching process, a high etch selectivity of a material of the insulating layer **106** to that of the dielectric layer **103** exists.

(61) In summary, with reference to FIG. 2, the electrical connection layer **105** covers the top surface of the third region III of the semiconductor channel **102** and extends to a partial sidewall of the semiconductor channel **102**, which therefore is beneficial to increasing the contact area between the electrical connection layer **105** and the semiconductor channel **102**, reducing the contact resistance between the electrical connection layer **105** and the third region III of the semiconductor channel **102**, and improving the transmission performance of electrical signals between the electrical connection layer **105** and the third region III of the semiconductor channel **102**, thereby improving electrical properties of the semiconductor structure. Further, this is beneficial to improving the transmission performance of electrical signals between the other conductive structures and the third region III of the semiconductor channel **102**, when the conductive structures of other functional devices are subsequently formed on the electrical connection layer **105**. Therefore, the embodiments of the present disclosure are beneficial to improving the ability of the electrical signal that the electrical connection layer **105** transmits to the semiconductor channel **102** to control the semiconductor channel **102**, thereby improving the on/off sensitivity of the GAA transistor.

(62) Another embodiment of the present disclosure further provides a method of manufacturing a semiconductor structure, which is used to manufacture the semiconductor structure in the above embodiment. FIG. 6 to FIG. 18 are schematic structural cross-sectional views corresponding to various steps of a method of manufacturing a semiconductor structure according to another embodiment of the present disclosure. The method of manufacturing a semiconductor structure provided by another embodiment is described in detail below with reference to the drawings.

(63) It should be noted that to conveniently describe and clearly illustrate the steps of the method of manufacturing a semiconductor structure, FIGS. 6 to 18 in this embodiment are all schematic partial structural diagrams of the semiconductor structure. In addition, parts that are the same as or corresponding to the foregoing embodiment are not described herein again.

(64) FIG. 7 is a schematic cross-sectional view of the structure in FIG. 6 along a first sectional direction AA1, and FIG. 8 is a schematic cross-sectional view of the structure in FIG. 6 along a second sectional direction BB1. It should be noted that, one or both of the schematic cross-sectional views along the first cross-sectional direction AA1 and the second cross-sectional

direction BB1 are set according to the needs of the presentation.

(65) With reference to FIGS. **6** to **18**, the method of manufacturing the semiconductor structure includes: providing a base **100**, where the base **100** includes bit lines **101** arranged at intervals and a semiconductor channel **102**, the bit lines **101** extend along the first direction X, the semiconductor channel **102** is located on a part of the top surface of the bit line **101**, and in the direction Z perpendicular to the top surface of the bit line **101**, the semiconductor channel **102** includes a first region I, a second region II and a third region III that are arranged sequentially; and forming a dielectric layer **103**, where the dielectric layer **103** is located between adjacent bit lines **101** and on a sidewall of the semiconductor channel **102**.

(66) It should be noted that both the first region I and the third region III may be used as the source or drain of the subsequently formed GAA transistor having the semiconductor channel **102**. A part of the second region II corresponds to the first gate layer **114** of the GAA transistor to be formed subsequently, and a part of the third region III corresponds to the second gate layer **124** of the GAA transistor to be formed subsequently.

(67) In some embodiments, the providing a base may include the steps: providing an initial base (not shown in the figure), where the initial first dielectric layer (not shown in the figure) extending along a first direction X is disposed in the initial base; with reference to FIGS. **6** to **8**, patterning the initial base and the initial first dielectric layer, to form the bit lines **101** arranged at intervals, the semiconductor channel **102**, and the initial first dielectric layer **163** located between adjacent bit lines **101**, where a top surface of the initial first dielectric layer **163** is not lower than that of the semiconductor channel **102**, the sidewall of the semiconductor channel **102**, a sidewall of the initial first dielectric layer **163**, and the partial top surface of the bit line **101** define a trench **107**, and the trench **107** extends along the second direction.

(68) It should be noted that, with reference to FIG. **6**, the initial base is the basis for forming the bit line **101** and the semiconductor channel **102**. When the initial base and the initial first dielectric layer **163** are patterned to form the bit line **101** and the semiconductor channel **102**, a substrate **110** is formed.

(69) In this embodiment, both the initial base and the initial first dielectric layer **163** may be formed through a self-aligned quadruple patterning (SAQP) process or a self-aligned double patterning (SADP) process.

(70) In some embodiments, doping treatment and annealing treatment may also be performed on the initial base, such that the initial base is doped with N-type doping elements or P-type doping elements. This is beneficial to improving the conductivity of the semiconductor channel **102** formed on the basis of the initial base, thereby reducing the turn-on voltage between the first region I and the third region III, that is, reducing the turn-on voltage between the source and the drain of the GAA transistor formed subsequently. In addition, the initial base is doped with N-type doping elements or P-type doping elements to improve the conductivity of the bit line **101** formed on the basis of the initial base, thereby reducing the contact resistance between the first region I and the bit line **101**, and improving the electrical property of the semiconductor structure.

(71) With reference to FIGS. **9** to **18**, a gate structure **104** is formed. The gate structure **104** surrounds at least the dielectric layer **103** in the second region II and extends along the second direction Y, and the first direction X is different from the second direction Y. An electrical connection layer **105** is formed. The electrical connection layer **105** covers the top surface of the third region III and extends to a partial sidewall of the semiconductor channel **102**. The electrical connection layer **105** and the dielectric layer **103** jointly cover the surface of the semiconductor channel **102**. An insulating layer **106** is formed and located between adjacent semiconductor channels **102** on the same bit line **101**.

(72) In some embodiments, with reference to FIG. **17**, the gate structure **104** includes a first gate layer **114** and a second gate layer **124**, and the step of forming the dielectric layer **103**, the gate structure **104**, and the insulating layer **106** may include the following steps.

(73) With reference to FIG. 10, a second dielectric layer **123** is formed on the sidewall of the trench **107** (with reference to FIG. 6) in the first region I. A first insulating layer **116** is formed and located in the trench **107**, and isolates the adjacent second dielectric layers **123**. The top surface of the first insulating layer **116** is not lower than the top surface of the semiconductor channel **102**. The top surface of the first insulating layer **116** is not lower than the top surface of the semiconductor channel **102**, which facilitates the subsequent formation of second gaps between the first insulating layer **116** and the semiconductor channels **102** in the second region II and the third region III. Then, the first gate layer and the second gate layer with precise dimensions can be formed in the second gaps through self-alignment. This is beneficial to simplifying the formation steps of the first gate layer and the second gate layer, and the size of the second gap is adjusted to obtain the first gate layer and the second gate layer in a small size.

(74) The step of forming the second dielectric layer **123** and the first insulating layer **116** may include the following steps.

(75) With reference to FIG. 9, a preliminary second dielectric layer **173** is formed on the sidewall of the trench **107** (with reference to FIG. 6), and a first gap is formed between adjacent preliminary second dielectric layers **173**. In some embodiments, the initial second dielectric layer **173** may be formed by using the following process step: performing a deposition process to form a surface covering the top surface of the semiconductor channel **102** and exposing all sidewalls, and further form a top surface and a sidewall exposed by the initial first dielectric layer **163** (with reference to FIG. 8). The material of the initial second dielectric layer **173** includes silicon oxide.

(76) With reference to FIG. 9, a first insulating layer **116** is formed in the first space. In some embodiments, the first insulating layer **116** may be formed by using the following process steps: forming a first insulating film covering the top surface of the initial second dielectric layer **173** and filling up the first space; and performing chemical mechanical planarization process on the first insulating film until the initial second dielectric layer **173** is exposed, where the retained first insulating layer is used as the first insulating layer **116**. The material of the first insulating layer **116** includes silicon nitride.

(77) The initial first dielectric layer **163** and the initial second dielectric layer **173** are made of a same material. This is beneficial to subsequently partially removing the initial first dielectric layer **163** and the initial second dielectric layer **173** in the same removal step, to form the second gap.

(78) In some embodiments, with reference to FIGS. 6 to 9, after the trench **107** is formed and before the second dielectric layer **123** (with reference to FIG. 10) is formed, metal silicidation is performed on a part of the top surface of the bit line **101** exposed by the trench **107** to form a metal-semiconductor compound structure **111**.

(79) With reference to FIG. 9, the metal-semiconductor compound structure **111** have a smaller resistivity than unmetallized semiconductor materials. Therefore, compared to the semiconductor channel **102**, the bit line **101** including the metal-semiconductor compound structure **111** has a smaller resistivity. This reduces the resistance of the bit lines **101** and the contact resistance between the bit lines **101** and the semiconductor channel **102** of the first region I, thereby further improving the electrical performance of the semiconductor structure.

(80) In some embodiments, with reference to FIG. 9, the step of performing metal silicidation on the part of the top surface of the bit line **101** exposed by the initial second dielectric layer **173** may include: forming a metal layer (not shown in the figure) is formed on the exposed top surface of the bit line **101**, and the metal layer provides metal elements for the metal-semiconductor compound structure **111**. The material of the metal layer includes at least one selected from the group consisting of cobalt, nickel, molybdenum, titanium, tungsten, tantalum and platinum.

(81) In other embodiments, the metal silicidation may not be performed on the exposed top surface of the bit line, and the second dielectric layer and the first insulating layer may be directly formed on the exposed top surface of the bit line.

(82) Then, with reference to FIGS. 9 to 11, the initial first dielectric layer **163** (with reference to

FIG. 8) and the initial second dielectric layer **173** are etched by using the first insulating layer **116** as a mask to form the first dielectric layer **113** and the second dielectric layer **123**.

(83) With reference to FIGS. **12** to **18**, a third dielectric layer **133** and a first gate layer **114** are formed on the sidewall of the trench **107** (with reference to FIG. **6**) in the second region II, where a top surface of the third dielectric layer **133** is higher than that of the first gate layer **114**. A fourth dielectric layer **143** is formed, and located between the first insulating layer **116** and the third dielectric layer **133**. A fifth dielectric layer **153** and a second gate layer **124** are formed on the sidewall of the trench **107** (with reference to FIG. **6**) in the third region III, where a top surface of the fifth dielectric layer **153** is higher than that of the second gate layer **124**. A second insulating layer **126** is formed and located between the first insulating layer **116** and the fifth dielectric layer **153**, where the insulating layer **106** includes the first insulating layer **116** and the second insulating layer **126**.

(84) In some embodiments, the step of forming the third dielectric layer **133**, the first gate layer **114**, and the second insulating layer **126** may include the following steps.

(85) With reference to FIGS. **12** to **14**, an initial third dielectric layer **183** is formed on the sidewalls of the semiconductor channels **102** in the second region II and the third region III, and a second gap **129** is formed between the initial third dielectric layer **183** and the first insulating layer **116**. In some embodiments, the initial third dielectric layer **183** may be formed by the following process steps: performing thermal oxidation treatment on the exposed surfaces of the semiconductor channels **102** in the second region II and the third region III to form the initial third dielectric layer **183**. The material of the initial third dielectric layer **183** is silicon oxide. In other embodiments, the initial third dielectric layer covering the surfaces of the semiconductor channels in the second region and the third region may further be formed through a deposition process.

(86) With reference to FIGS. **13** and **14**, the first gate layer **114** is formed in the second gap **129** of the second region II. The step of forming the first gate layer **114** may include: with reference to FIG. **12**, forming an initial first gate layer **134**, where the initial first gate layer **134** fills up the second gap **129** and is located on the top surface of the initial third dielectric layer **183**; and with reference to FIG. **13**, removing, through etching, the initial first dielectric layer **134** surrounding the sidewall of the semiconductor channel **102** in the third region III, partially surrounding the sidewall of the semiconductor channel **102** in the second region II, and located on the top surface of the initial third dielectric layer **183**, where the remaining initial first gate layer **134** is used as the first gate layer **114**, and the first gate layer **114** only partially surrounds the sidewall of the semiconductor channels **102** in the second region II.

(87) With reference to FIG. **15**, an initial fourth dielectric layer **193** is formed in the remaining second spaces **129** (with reference to FIG. **13**). In some embodiments, the initial fourth dielectric layer **193** may be formed by using the following process steps: performing a deposition process to form the initial fourth dielectric layer **193** that fills up the remaining second gap **129** and covers the top surface of the initial third dielectric layer **183**; and performing chemical mechanical polishing on the initial fourth dielectric layer **193** until the initial third dielectric layer **183** is exposed. The material of the initial fourth dielectric layer **193** includes silicon oxide.

(88) With reference to FIGS. **15** to **16**, the first insulating layer **116** is used as a mask to etch the initial third dielectric layer **183** and the initial fourth dielectric layer **193**, to form the third dielectric layer **133** and the fourth dielectric layer **143**.

(89) In some embodiments, the step of forming the fifth dielectric layer **153**, the second gate layer **124**, and the second insulating layer **126** may include the following steps.

(90) With reference to FIG. **16**, the fifth dielectric layer **153** is formed on the sidewall of the third region III, and a third gap **139** is formed between the fifth dielectric layer **153** and the first insulating layer **116**. A second gate layer **124** is formed in a part of the third gap **139**. In some examples, the fifth dielectric layer **153** is further formed on the top surface of the third region III, and the fifth dielectric layer **153** may be formed by using the following process steps: thermally

oxidizing the surface of the semiconductor channel **102** in the third region III to form the fifth dielectric layers **153**. The material of the fifth dielectric layer **153** is silicon oxide. In other embodiments, the initial fifth dielectric layer covering the surface of the semiconductor channel in the third region may further be formed through a deposition process.

(91) With reference to FIG. **16**, the step of forming the second gate layer **124** may include: forming an initial second gate layer (not shown in the figure), where the initial second gate layer fills up the third gap **139** and is located on the top surface of the fifth dielectric layer **153**; and etching the initial second gate layer partially surrounding the sidewall of the semiconductor channel **102** in the third region and located on the top surface of the fifth dielectric layer **153**, where the retained initial second gate layer is used as the second gate layer **124**, and the second gate layer **124** only partially surrounds the sidewall of the semiconductor channel **102** in the third region III.

(92) With reference to FIG. **17**, the second insulating layer **126** is formed in the remaining third gap **139** (with reference to FIG. **16**). In some embodiments, the following process steps may be used to form the second insulating layer **126**: performing a deposition process, to form a second insulating film that fills up the retained third gap **139** and covers the top surface of the fifth dielectric layer **153**; and performing chemical mechanical polishing on the second insulating film and the first insulating layer **116** until the fifth dielectric layer **153** is exposed, where the third insulating film is used as the second insulating layer **126**. The material of the second insulating layer **126** may be silicon nitride.

(93) In some embodiments, with reference to FIG. **17**, after the second gate layer **124** is formed and before the second insulating layer **126** is formed, the manufacturing method may further include: doping at least one of lanthanide elements and/or zirconium into the second gate layer **124**.

(94) The doping at least one of lanthanide elements and/or zirconium into the second gate layer **124** may be performed through the following process: with reference to FIG. **16**, forming a first diffusion layer (not shown in the figure) in the retained third gap **139**, increasing the temperature of the environment in which the semiconductor structure is located, and doping metal elements into the second gate layer **124** through thermal diffusion of metal elements in the first diffusion layer. It should be noted that the doping depth of the metal elements in the second gate layer **124** increases as the temperature of the environment where the semiconductor structure is located increases. The material of the first diffusion layer may be at least one of lanthanide oxides such as lanthanum oxide, scandium oxide, or ceria or zirconium oxide.

(95) After the process of doping at least one of the lanthanides and/or the zirconium element into the second gate layer **124** ends, the remaining diffusion layer is removed to expose the second gate layer **124** in preparation for the subsequent formation of the second insulating layer **126** on the top surface of the second gate layer **124**.

(96) In some embodiments, with reference to FIGS. **3**, **4**, and **18**, the step of forming the electrical connection layer **105** may include: etching the fifth dielectric layer **153** by using the insulating layer **106** as a mask, to expose the top surface and a partial sidewall of the third region III, and form a groove **117** between the insulating layer **106** and the third region III; and forming the electrical connection layer **105** filling up the groove **117**.

(97) The electrical connection layer **105** may be formed by using the following process steps: performing a deposition process to form a conductive film that fills up the groove **117** and covers the top surface of the insulating layer **106**; performing chemical mechanical polishing on the conductive film until the insulating layer **106** is exposed, where the remaining conductive film is used as the electrical connection layer **105**.

(98) In some embodiments, the method of manufacturing the semiconductor structure may further include: with reference to FIG. **5**, doping at least one of lanthanide elements and/or zirconium into the electrical connection layer **105**. The doping at least one of lanthanide elements and/or zirconium into the electrical connection layer **105** may be performed through the following process: forming a second diffusion layer (not shown in the figure) on the top surface of the

electrical connection layer **105**, increasing the temperature of the environment in which the semiconductor structure is located, and doping metal elements into the electrical connection layer **105** through thermal diffusion of metal elements in the first diffusion layer. It should be noted that the doping depth of the metal elements in the electrical connection layer **105** increases as the temperature of the environment where the semiconductor structure is located increases. The material of the second diffusion layer may be at least one of lanthanide oxides such as lanthanum oxide, scandium oxide, or ceria or zirconium oxide.

(99) At least one of the lanthanides and/or zirconium are/is doped into the electrical connection layer **105**. In this way, the electrical performance of the electrical connection layer **105** is improved, thereby improving the transmission effect of electrical signals in the electrical connection layer **105**, to improve the ability of the electrical signal transmitted to the semiconductor channel **102** through the electrical connection layer **105** to control the semiconductor channel **102**. This is beneficial to improving the on/off sensitivity of the GAA transistor.

(100) It should be noted that, in other embodiments, after the first gate layer is formed, the second insulating layer and the electrical connection layer may be formed directly on the top surface of the first gate layer, that is, the gate structure only includes the first gate layer.

(101) In some embodiments, the method of manufacturing the semiconductor structure may further include: forming a conductive structure of another functional device on the top surface of the electrical connection layer **105** away from the bit line **101**.

(102) In summary, with reference to FIG. **3**, the electrical connection layer **105** covering the top surface of the third region III of the semiconductor channel **102** and extending to a partial sidewall of the semiconductor channel **102** is formed, which therefore is beneficial to increasing the contact area between the electrical connection layer **105** and the semiconductor channel **102**, reducing the contact resistance between the electrical connection layer **105** and the third region III of the semiconductor channel **102**, and improving the transmission performance of electrical signals between the electrical connection layer **105** and the third region III of the semiconductor channel **102**, thereby improving electrical properties of the semiconductor structure. Further, this is beneficial to improving the transmission performance of electrical signals between the other conductive structures and the third region III of the semiconductor channel **102**, when the conductive structures of other functional devices are subsequently formed on the electrical connection layer **105**. Therefore, from the above two aspects, it can be seen that the electrical connection layer **105** transmits electrical signals to the semiconductor channel **102**, which is beneficial to improving the ability of the electrical signal to control the semiconductor channel **102**, thereby improving the on/off sensitivity of the GAA transistor.

(103) The embodiments or implementations of this specification are described in a progressive manner, and each embodiment focuses on differences from other embodiments. The same or similar parts between the embodiments may refer to each other.

(104) In the description of this specification, the description with reference to terms such as “an embodiment”, “an exemplary embodiment”, “some implementations”, “a schematic implementation”, and “an example” means that the specific feature, structure, material, or characteristic described in combination with the implementation(s) or example(s) is included in at least one implementation or example of the present disclosure.

(105) In this specification, the schematic expression of the above terms does not necessarily refer to the same implementation or example. Moreover, the described specific feature, structure, material or characteristic may be combined in an appropriate manner in any one or more implementations or examples.

(106) It should be noted that in the description of the present disclosure, the terms such as “center”, “top”, “bottom”, “left”, “right”, “vertical”, “horizontal”, “inner” and “outer” indicate the orientation or position relationships based on the accompanying drawings. These terms are merely intended to facilitate description of the present disclosure and simplify the description, rather than

to indicate or imply that the mentioned apparatus or element must have a specific orientation and must be constructed and operated in a specific orientation. Therefore, these terms should not be construed as a limitation to the present disclosure.

(107) It can be understood that the terms such as “first” and “second” used in the present disclosure can be used to describe various structures, but these structures are not limited by these terms. Instead, these terms are merely intended to distinguish one structure from another.

(108) The same elements in one or more accompanying drawings are denoted by similar reference numerals. For the sake of clarity, various parts in the accompanying drawings are not drawn to scale. In addition, some well-known parts may not be shown. For the sake of brevity, a structure obtained by implementing a plurality of steps may be shown in one figure. In order to understand the present disclosure more clearly, many specific details of the present disclosure, such as the structure, material, size, processing process, and technology of the device, are described below. However, as those skilled in the art can understand, the present disclosure may not be implemented according to these specific details.

(109) Finally, it should be noted that the above embodiments are merely intended to explain the technical solutions of the present disclosure, rather than to limit the present disclosure. Although the present disclosure is described in detail with reference to the above embodiments, those skilled in the art should understand that they may still modify the technical solutions described in the above embodiments, or make equivalent substitutions of some or all of the technical features recorded therein, without deviating the essence of the corresponding technical solutions from the scope of the technical solutions of the embodiments of the present disclosure.

INDUSTRIAL APPLICABILITY

(110) In a semiconductor structure and a manufacturing method thereof provided by the embodiments of the present disclosure, vertical GAA transistors are formed in a base, and bit lines are buried in the base and located under the semiconductor channel. Therefore, a 3D stacked semiconductor structure can be formed, and the transistors are arranged in a 4F_{sup.2} manner to improve the integration density of the semiconductor structure. In addition, the electrical connection layer and the dielectric layer cover the surface of the semiconductor channel. The electrical connection layer not only covers the top surface of the third region of the semiconductor channel but also extends to a partial sidewall of the semiconductor channel, which therefore is beneficial to increasing the contact area between the electrical connection layer and the semiconductor channel, reducing the contact resistance between the electrical connection layer and the third region of the semiconductor channel, and improving the transmission performance of electrical signals between the electrical connection layer and the third region of the semiconductor channel, thereby improving electrical properties of the semiconductor structure. Further, this is beneficial to improving the transmission performance of electrical signals between the other conductive structures and the third region of the semiconductor channel, when the conductive structures of other functional devices are subsequently formed on the electrical connection layer. Therefore, the embodiments of the present disclosure are beneficial to improving the ability of the electrical signal that the electrical connection layer transmits to the semiconductor channel to control the semiconductor channel, thereby improving the on/off sensitivity of the GAA transistor.

Claims

1. A semiconductor structure, comprising: a base, wherein the base comprises: bit lines arranged at intervals and extending along a first direction, and a semiconductor channel located on partial top surfaces of the bit lines, and along a direction from the bit line to the semiconductor channel, the semiconductor channel comprises a first region, a second region, and a third region that are arranged sequentially; a dielectric layer located between adjacent two of the bit lines and on a sidewall of the semiconductor channel; a gate structure at least surrounding the dielectric layer in

the second region and extending along a second direction, wherein the first direction is different from the second direction; an electrical connection layer covering a top surface of the third region and extending to a partial sidewall of the semiconductor channel, wherein the electrical connection layer and the dielectric layer cover a surface of the semiconductor channel; and an insulating layer located between adjacent two of the semiconductor channels on a same bit line and isolating the gate structure as well as the electrical connection layer on adjacent two of the dielectric layers; wherein the gate structure comprises: a first gate layer surrounding the dielectric layer in the second region and extending along the second direction; and a second gate layer surrounding the dielectric layer in the third region, wherein in a direction perpendicular to a top surface of the bit line, the second gate layer and the first gate layer are disposed at intervals; wherein along the direction from the bit line to the semiconductor channel, a ratio of a height of the first gate layer surrounding a sidewall of the second region to a height of the semiconductor channel in the second region is $1/2$ to $9/10$.

2. The semiconductor structure according to claim 1, wherein the electrical connection layer comprises at least one of lanthanides or zirconium.

3. The semiconductor structure according to claim 1, wherein along the direction from the bit line to the semiconductor channel, the electrical connection layer on the top surface of the third region is 10 nm to 50 nm.

4. The semiconductor structure according to claim 1, wherein along a direction perpendicular to the sidewall of the semiconductor channel, a ratio of a width of the electrical connection layer on the sidewall of the semiconductor channel to a width of the semiconductor channel in the third region is $1/5$ to $3/5$.

5. The semiconductor structure according to claim 1, wherein along the direction from the bit line to the semiconductor channel, a ratio of a height of the electrical connection layer on the sidewall of the semiconductor channel to a height of the semiconductor channel in the third region is $1/5$ to $3/5$.

6. The semiconductor structure according to claim 1, wherein an orthographic projection of the electrical connection layer on the bit line is not overlapped or partially overlapped with an orthographic projection of the gate structure on the bit line.

7. The semiconductor structure according to claim 1, wherein in a plane perpendicular to the sidewall of the semiconductor channel, a cross-sectional area of the semiconductor channel in the first region, a cross-sectional area of the semiconductor channel in the second region, and a cross-sectional area of the semiconductor channel in the third region are decreased sequentially.

8. The semiconductor structure according to claim 1, wherein along the first direction, a side surface of the second gate layer away from the sidewall of the semiconductor channel is flush with a side surface of the first gate layer away from the sidewall of the semiconductor channel.

9. The semiconductor structure according to claim 1, wherein in a direction perpendicular to the sidewall of the semiconductor channel, the first gate layer is thinner than the second gate layer.

10. The semiconductor structure according to claim 1, wherein along the direction from the bit line to the semiconductor channel, a ratio of a height of the first gate layer surrounding a sidewall of the second region to a height of the semiconductor channel in the second region is $1/2$ to $9/10$.

11. The semiconductor structure according to claim 1, further comprising a metal-semiconductor compound structure at least located in a bit line opposite to a bottom of the insulating layer.

12. The semiconductor structure according to claim 11, wherein along a direction from the semiconductor channels located at two sides of the insulating layer to the insulating layer, the metal-semiconductor compound structure becomes deeper.

13. A method of manufacturing a semiconductor structure, comprising: providing a base, wherein the base comprises: bit lines arranged at intervals and extending along a first direction, and a semiconductor channel located on partial top surfaces of the bit lines, and along a direction perpendicular to the top surface of the bit line, the semiconductor channel comprises a first region,

a second region, and a third region that are arranged sequentially; forming a dielectric layer located between adjacent two of the bit lines and on a sidewall of the semiconductor channel; forming a gate structure at least surrounding the dielectric layer in the second region and extending along a second direction, wherein the first direction is different from the second direction; forming an electrical connection layer covering a top surface of the third region and extending to a partial sidewall of the semiconductor channel, wherein the electrical connection layer and the dielectric layer cover a surface of the semiconductor channel; and forming an insulating layer located between adjacent two of the semiconductor channels on a same bit line; wherein the gate structure comprises: a first gate layer surrounding the dielectric layer in the second region and extending along the second direction; and a second gate layer surrounding the dielectric layer in the third region, wherein in a direction perpendicular to a top surface of the bit line, the second gate layer and the first gate layer are disposed at intervals; wherein along the direction from the bit line to the semiconductor channel, a ratio of a height of the first gate layer surrounding a sidewall of the second region to a height of the semiconductor channel in the second region is $1/2$ to $9/10$.

14. The method of manufacturing according to claim 13, wherein the providing a base comprises: providing an initial base, wherein an initial first dielectric layer extending along the first direction is disposed in the initial base; and patterning the initial base and the initial first dielectric layer, to form the bit lines arranged at intervals, the semiconductor channel, and the initial first dielectric layer located between adjacent bit lines, wherein a top surface of the initial first dielectric layer is not lower than a top surface of the semiconductor channel, the sidewall of the semiconductor channel, a sidewall of the initial first dielectric layer, and the partial top surface of the bit line define a trench, and the trench extends along the second direction.

15. The method of manufacturing according to claim 14, wherein the forming the dielectric layer, the gate structure, and the insulating layer comprise: forming a second dielectric layer on a sidewall of the trench in the first region; forming a first insulating layer, wherein the first insulating layer is located in the trench and isolates adjacent two of the second dielectric layers, and a top surface of the first insulating layer is not lower than the top surface of the semiconductor channel; forming a third dielectric layer and a first gate layer on the sidewall of the trench in the second region, wherein a top surface of the third dielectric layer is higher than a top surface of the first gate layer; forming a fourth dielectric layer located between the first insulating layer and the third dielectric layer; forming a fifth dielectric layer and a second gate layer on the sidewall of the trench in the third region, wherein a top surface of the fifth dielectric layer is higher than a top surface of the second gate layer; and forming a second insulating layer located between the first insulating layer and the fifth dielectric layer, wherein the insulating layer comprises the first insulating layer and the second insulating layer.

16. The method of manufacturing according to claim 15, wherein after the trench is formed, and before the second dielectric layer is formed, metal silicidation is performed on a partial top surface of the bit line exposed by the trench, to form a metal-semiconductor compound structure.

17. The method of manufacturing according to claim 15, wherein the forming an electrical connection layer comprises: etching the fifth dielectric layer by using the insulating layer as a mask, until the top surface and a partial sidewall of the third region are exposed, to form a groove between the insulating layer and the third region; and forming the electrical connection layer filling up the groove.

18. The method of manufacturing according to claim 13, further comprising: doping at least one of lanthanides or zirconium into the electrical connection layer.
